SPETRO GEOPORTAL ACADEMY | OFFICIAL COURSE HANDBOOK

# SPETRO Geoportal Practical Field Manual & Workflows

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Scope: Practical Field Manual & Reference Guide | Pages: 10

## Course Overview & Objectives

Step-by-step guide to using the SPETRO platform for flood risk modeling, habitat suitability, watershed delineation, and satellite index processing.

MODULE OUTLINE:


- **1. SPETRO Platform Architecture**
- **2. Flood Hazard Assessment & Modeling**
- **3. Habitat Suitability & Land Suitability (AHP)**
- **4. Water Harvesting & Runoff Yield Modeling**
- **5. High-Availability Operational Guide**


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# 1. SPETRO Platform Architecture

Hybrid cloud deployment: FastAPI backend, Cloudflare Edge Worker reverse proxy, Google Earth Engine microservices, and Hugging Face persistent open storage.

### KEY TAKEAWAYS & PRACTICAL LAB INSTRUCTIONS

1. Load corresponding dataset layers from the SPETRO Data Harvester or Hugging Face.
2. Execute geoprocessing algorithms and verify outputs in the Interactive Map Editor.
3. Export cartographic layouts, GeoTIFF rasters, or GeoJSON vector polygons for field use.

## 2. Flood Hazard Assessment & Modeling

Rainfall-runoff modeling, HEC-HMS integration, topographic wetness index (TWI), DEM hydro-enforcement, and critical infrastructure vulnerability analysis.

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## 3. Habitat Suitability & Land Suitability (AHP)

Multi-criteria decision analysis (MCDA), Analytic Hierarchy Process (AHP) pair-wise comparison matrices, consistency index (CI) verification, and weighted overlay rasters.

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## 4. Water Harvesting & Runoff Yield Modeling

SCS-CN curve number runoff calculation, catchment area delineation, roof harvesting potentials, and seasonal reservoir sizing for East African communities.

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## 5. High-Availability Operational Guide

Automated service account rotation, Cloudflare edge tile caching, sub-second API responses, and resilient offline capabilities.

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